

ICT2217 Network Security
Lab Exercise 1.2: Switch Spoofing Attack and Defense

Learning Outcomes

Upon completion of this laboratory exercise, you should be able to:

- Conduct and defend against switch spoofing attack

Required Hardware

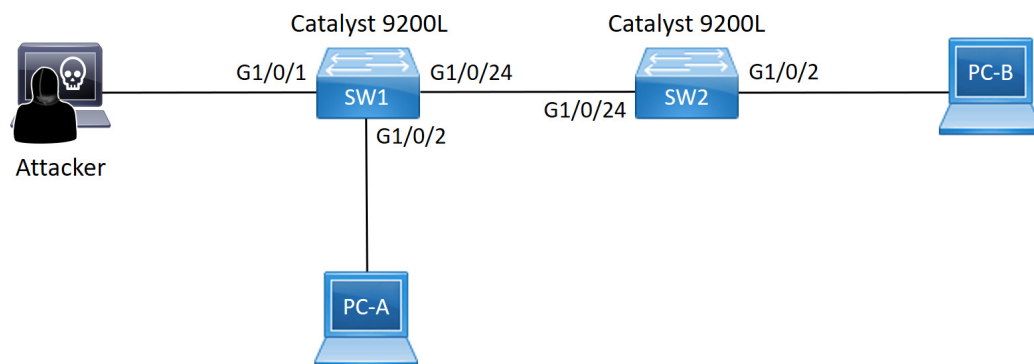
- 1 x Rack of Cisco networking devices
- 1 x Box of cables containing
 - USB-to-DB9 serial adapter + DB9-to-RJ45 console cable
 - USB-to-RJ45 console adapter
 - Ethernet cables
- 3 x Laptops

Required Software

- Tera Term <https://teratermproject.github.io/index-en.html>
- Kali Linux Live Boot on USB drive

PART 1: Conducting Switch Spoofing Attack using Yersinia

1.1 Set up the network topology as shown:

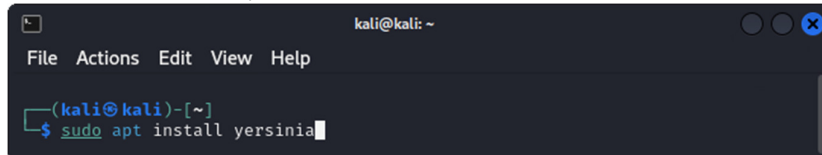


- 1.2 Configure interface G1/0/24 connecting switches SW1 and SW2 as trunk, and interface G1/0/2 connecting PC-A and PC-B as VLAN 10. Leave SW1 G1/0/1 connecting to Attacker's PC in default VLAN 1 and default switchport mode dynamic auto.
- 1.3 Boot up PC-A, PC-B and Attacker's PC in Kali Linux.
- 1.4 Configure suitable IP addresses on PC-A and PC-B manually. No requirement to configure IP address for Attacker's PC.

- 1.5 Verify that PC-A is able to ping PC-B to ensure that your setup is working correctly.
- 1.6 Before commencing switch spoofing attack, verify the mode of the interface connecting the attacker's PC.

```
SW1# sh int g1/0/1 trunk
```

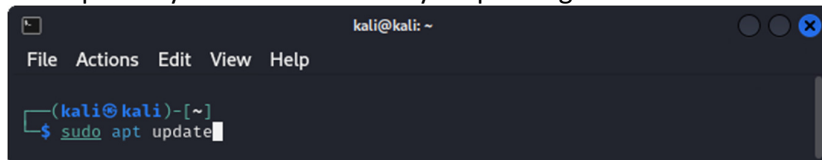
- 1.7 At the attacker's PC, start a terminal and install Yersinia as shown:



```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo apt install yersinia
```

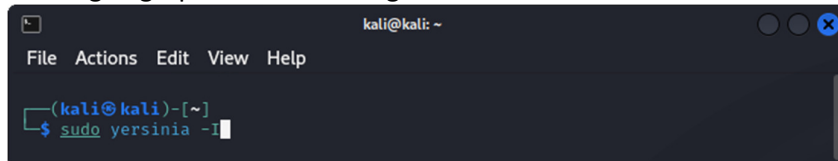
Note: You will need to login to ICT-WIFI to access the Internet as SIT-WIFI will block Kali sites.

- 1.8 If you encounter unable to locate package error message, update the package list from the repository as shown and re-try Step 1.7 again.



```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo apt update
```

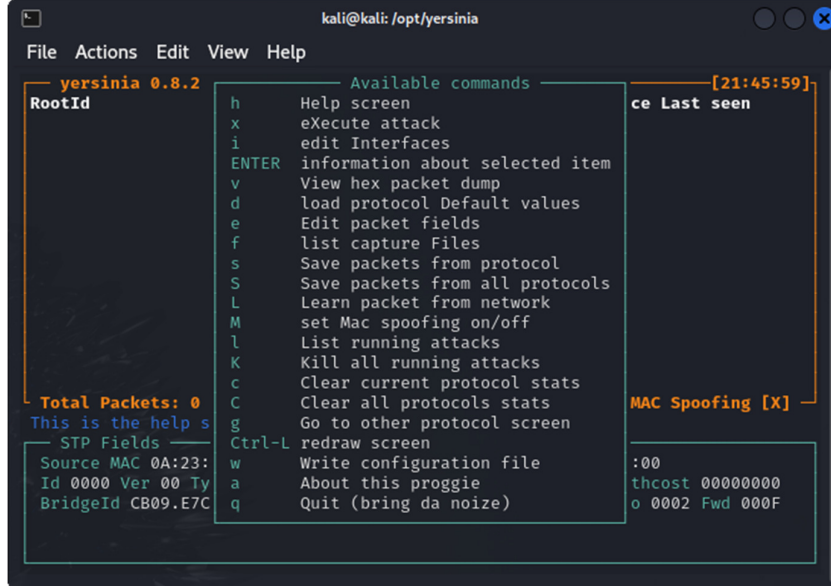
- 1.9 Once Yersinia is installed successfully, run it in interactive mode as there is error running in graphical mode using Kali 2025.1c version.



```
kali@kali: ~  
File Actions Edit View Help  
(kali@kali)-[~]  
$ sudo yersinia -i
```

WARNING: Logout and disable Wi-Fi to prevent attacking other networks through Wi-Fi which may get you into trouble.

1.10 Press key 'h' for help to see the list of commands available.



```

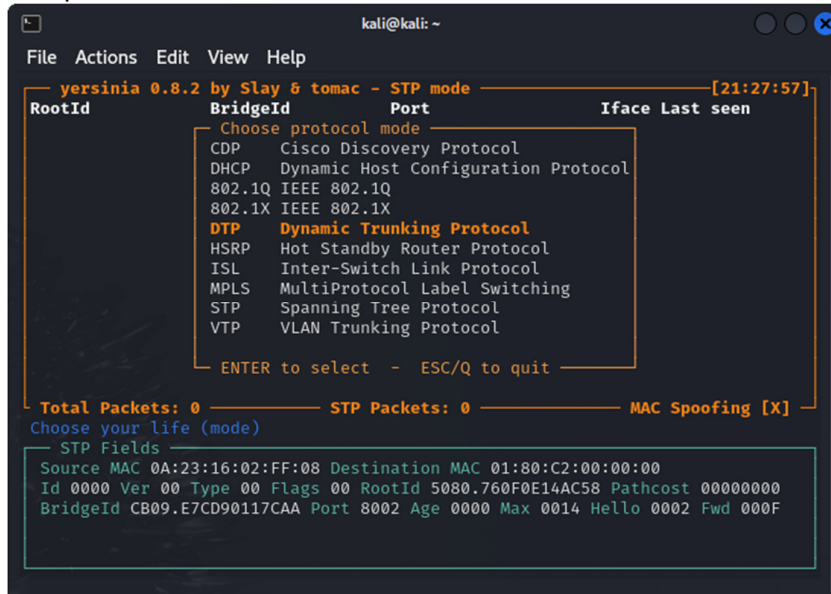
kali@kali: /opt/yersinia
File Actions Edit View Help

yersinia 0.8.2
RootId
Total Packets: 0
This is the help s
STP Fields
Source MAC 0A:23:
Id 0000 Ver 00 Ty
BridgeId CB09.E7C

Available commands
h Help screen
x eXecute attack
i edit Interfaces
ENTER information about selected item
v View hex packet dump
d load protocol Default values
e Edit packet fields
f list capture Files
s Save packets from protocol
S Save packets from all protocols
L Learn packet from network
M set Mac spoofing on/off
l List running attacks
K Kill all running attacks
c Clear current protocol stats
C Clear all protocols stats
g Go to other protocol screen
Ctrl-L redraw screen
w Write configuration file
a About this proggie
q Quit (bring da noize)

[21:45:59]
ce Last seen
MAC Spoofing [X]
:00
thcost 00000000
o 0002 Fwd 000F
  
```

1.11 Press key 'g' to go to protocol screen, use arrow keys to navigate and ENTER to select DTP protocol mode.



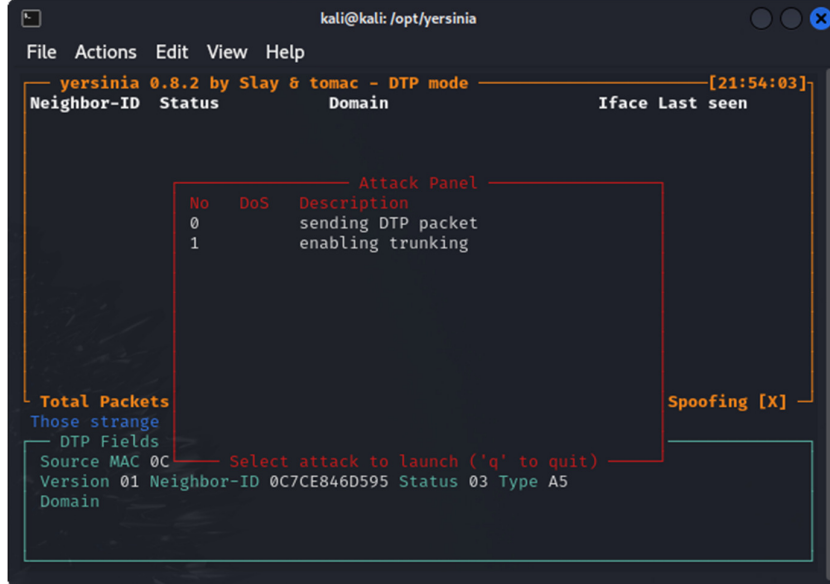
```

kali@kali: ~
File Actions Edit View Help

yersinia 0.8.2 by Slay & tomac - STP mode
RootId BridgeId Port Iface Last seen
Choose protocol mode
CDP Cisco Discovery Protocol
DHCP Dynamic Host Configuration Protocol
802.1Q IEEE 802.1Q
802.1X IEEE 802.1X
DTP Dynamic Trunking Protocol
HSRP Hot Standby Router Protocol
ISL Inter-Switch Link Protocol
MPLS MultiProtocol Label Switching
STP Spanning Tree Protocol
VTP VLAN Trunking Protocol
ENTER to select - ESC/Q to quit

Total Packets: 0 STP Packets: 0 MAC Spoofing [X]
Choose your life (mode)
STP Fields
Source MAC 0A:23:16:02:FF:08 Destination MAC 01:80:C2:00:00:00
Id 0000 Ver 00 Type 00 Flags 00 RootId 5080.760F0E14AC58 Pathcost 00000000
BridgeId CB09.E7CD90117CAA Port 8002 Age 0000 Max 0014 Hello 0002 Fwd 000F
  
```

1.12 Press key 'x' to eXecute attack panel.



```

kali@kali: /opt/yersinia
File Actions Edit View Help
yersinia 0.8.2 by Slay & tomac - DTP mode [21:54:03]
Neighbor-ID Status Domain Iface Last seen

Attack Panel
No DoS Description
0 sending DTP packet
1 enabling trunking

Total Packets
Those strange
DTP Fields
Source MAC 0C
Version 01 Neighbor-ID 0C7CE846D595 Status 03 Type A5
Domain

Select attack to launch ('q' to quit)
Spoofing [X]
  
```

1.13 Finally, press key '1' enabling trunking to launch DTP attack.

1.14 After the attack, show the mode of the interface connecting the attacker's PC again. What difference do you observe as compared to Step 1.6?

1.15 At the attacker's PC, start Wireshark.

1.16 From PC-A, ping PC-B. Can your Wireshark capture the ping packets? Why or why not? Explain.

1.17 If you could not capture the ping packets in Step 1.16, troubleshoot the problem so that you are able to at least capture one ping packet successfully.

1.18 Are you able to enhance your attack so that you are able to capture all ping packets successfully?

PART 2: Defending against Switch Spoofing Attack

2.1 Implement defense against switch spoofing attack as discussed.

2.2 Launch Yersinia to conduct switch spoofing attack again.

2.3 Show the mode of the interface connected to the attacker's PC after launching DTP attack. Do you observe any difference?

2.4 Do you think the recommended defenses are effective against switch spoofing attack? Explain.